

Andrew Stirling



TECHNICAL SKILLS

Programming: C++, Python, CMake, Matlab

Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, Git, Docker, Foxglove, Gazebo, MuJoCo

Research Areas: SLAM, Visual-Inertial Odometry, UWB Localization, Sensor Fusion & Calibration, Lie-Group Theory, Probabilistic Modelling, Bayesian Estimation, 3D Scene Representations.

Languages: English (Native), French (C1)

EXPERIENCE

M.Sc. Thesis Candidate

Sept. 2024 – Present

DECAR Group, Supervised by Prof. James R. Forbes

Montréal, CAN

- Derived a novel ML-based variational inference framework for robotic localization handling non-line-of-sight (NLOS) and multipath effects enabling improved geospatial sensing.
- Developed a C++/ROS package for UWB-visual-inertial SLAM for GPS-denied autonomous navigation.
- Performed statistical data analysis on large experimental datasets to evaluate estimator accuracy, consistency, and convergence behaviour.
- Authored and presented peer-reviewed work in *IEEE Robotics and Automation Letters (RA-L)* and at *ICRA*, demonstrating strong written and oral communication of complex math and estimation techniques.

Co-Founder

Sept. 2023 – Aug. 2024

OPSIS

Montréal, CAN

- Capstone design project turned startup accepted into the McGill Dobson TechAccel incubator program.
- Designed and implemented Kalman filtering-based state estimation algorithms for occlusion-robust tool localization.
- Collaborated across software, mechanical, and clinical teams to translate mathematical algorithms into a deployable system.

Robotics Research Intern

Jun. 2023 – Aug. 2023

University of Applied Sciences & Technology

Berlin, GER

- Conducted extensive testing of diverse quadruped control strategies utilizing MuJoCo and validated outcomes through physical testing on the Unitree Go1 Robot, utilizing a ROS2 package tailored for low-level control.
- Assisted in the development and successful implementation of an Iterative Learning Controller (ILC), capable of tracking commanded velocities within 5 % relative error.
- Co-authored paper detailing control framework accepted and presented at the European Control Conference (ECC).

EDUCATION

M.Sc Mechanical Engineering (Robotics & Autonomous Systems)

Sept. 2024 – May 2026

McGill University

Montréal, CAN

- *GPA:* 4.0/4.0.
- *Thesis:* Ultra-wideband sensing for autonomous navigation in GPS-denied environments.
- *Honors & Awards:* Ian McLachlin Prize for Entrepreneurship in Engineering, McGill Engineering Undergraduate Student Master's Award

B.Eng, Mechanical Engineering (Minor Computer Science)

Sept. 2019 – May 2024

McGill University

Montréal, CAN

- *GPA:* 3.8/4.0.
- *Honors & Awards:* Engine Design Prize, Antje Graupe Pryor Award, RISE Globalink Award, Dean's Honour List, Alexander Rutherford Scholarship.
- *Extracurriculars:* McGill Men's Volleyball Team, McGill Robotics Club, McGill Tutor, and Teaching Assistant.

SELECTED PROJECTS

ESGVI [🏆★4] *RA-L / ICRA 2026* | Sparse Lie-group variational inference for state estimation, providing a scalable, uncertainty-aware alternative to traditional SLAM solvers.

UVIO [🏆★1] *TBA* | Estimation framework integrating visual, inertial, and UWB sensors for real-time drift-resilient navigation.

PUBLICATIONS

- [1] **Andrew Stirling**, Mykola Lukashchuk, et al. "Gaussian Variational Inference with Non-Gaussian Factors for State Estimation: A UWB Localization Case Study". In: *IEEE Robotics and Automation Letters* 11.3 (Mar. 2026), pp. 2762–2769.
- [2] Manuel Weiss, **Andrew Stirling**, et al. "Achieving Velocity Tracking Despite Model Uncertainty for a Quadruped Robot with a PD-ILC Controller". In: *2024 European Control Conference (ECC)*. June 2024, pp. 134–140.